K-stability

Let $M \hookrightarrow \mathbb{CP}^N$ be normal

$$\mathcal{O}(1)|_{M_{reg}} = K_{M_{reg}}^{-r}$$

Let $\lambda : \mathbb{C}^* \hookrightarrow SL(N+1, \mathbb{C})$

1-parameter subgroup s.t. $\lambda(S^1) \subset SU(N+1)$

<u>Definition</u> λ defines a <u>special degeneration</u> if $Y = \lim_{t \to 0} \lambda(t) \cdot M$ is normal

λ generated by matrix $a_{ij} \in su(N+1)$

Let $\theta_\lambda : \mathbb{CP}^N \to \mathbb{R}$

$$Z \longmapsto \sqrt{-1}\, \frac{a_{ij} Z_i \bar{Z}_j}{|Z|^2}$$

$\left(\lambda|_{(0,\infty)} \text{ is flow of } \nabla\theta_\lambda \right)$

<u>Definition</u> Donaldson-Futaki invariant of λ is

$$DF(M, \lambda) = -\frac{1}{V} \int_Y \theta_\lambda \left(n\,Ric(\omega) - \hat{R}\omega \right) \wedge \omega^{n-1}$$

$\omega = \omega_{FS}|_Y$, $\hat{R}$ = average scalar curvature of ω

V = volume of (Y, ω).

<u>Definition</u> M is <u>K-stable</u> if

$DF(M,\lambda)$ is non-negative $\forall$ special deg. λ

$DF(M,\lambda) = 0$ only if $Y \simeq M$

(allowing any r)

<u>Theorem</u> (Chen - Donaldson - Sun)

$\underset{\text{(smooth)}}{M}$ is K-stable $\iff$ it admits a KE metric

($\Leftarrow$ due to Tian)

<u>Proof sketch</u>

Suppose we can find ω_t on M s.t.

$$Ric(\omega_t) = t\omega_t + (1-t)\alpha$$

for all $t < 1$.

Then we have a sequence of embeddings

$$\Phi_k : M \longrightarrow \mathbb{CP}^N$$

using sections of K_M^{-m}

s.t. $\Phi_k(M) \longrightarrow Y \subset \mathbb{CP}^N$ normal variety

and Y admits a (singular) KE metric

<u>Claim</u> 1) $\mathrm{Aut}(Y)$ is reductive

2) $DF(Y, \lambda) = 0$

$\forall\, \lambda : \mathbb{C}^* \hookrightarrow \mathrm{Aut}(Y)$

<u>Reason</u> in case Y is smooth:

1) If $\mathrm{Ric}(\omega) = \omega$ on Y

then $\mathrm{Lie}\,\mathrm{Aut}(Y) = \left\{ f \;\middle|\; \Delta f + f = 0 \right\}$
$f : Y \to \mathbb{C}$

If $\mathrm{Re}\, f = 0$, then $\mathrm{Re}\, \nabla^{1,0} f$ generates an isometry. Therefore

$\mathrm{Aut}(Y)$ = complexification of $\mathrm{Isom}(Y, \omega)$

2) $DF(Y, \lambda)$ is independent of the metric on Y used to compute it

Using the KE metric, $DF(Y, \lambda) = 0$

If Y is singular, 1) and 2) follow from convexity of the Ding functional

[defined on metrics $e^{-\varphi}$ on $K_{Y_{reg}}^{-1}$ with bounded potentials by

$$D(\varphi) = -E(\varphi) - \log \int_{Y_{reg}} e^{-\varphi}$$

E = Monge-Ampère energy:

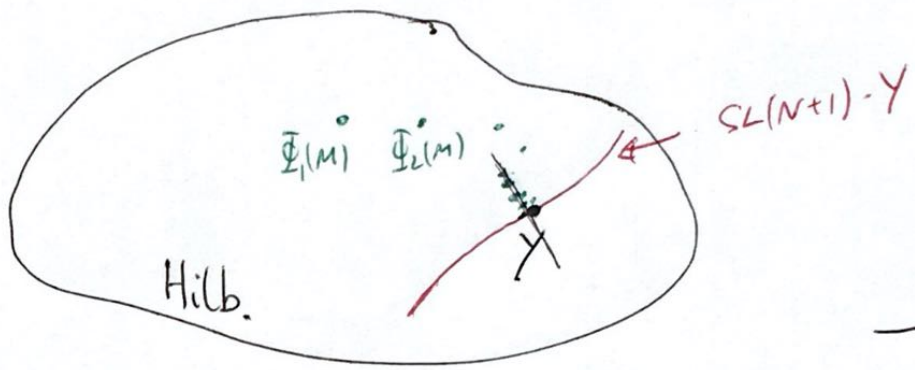
$\delta E(\varphi) = \frac{1}{V} \int \delta\varphi\, \omega_\varphi^n$, critical pts of D are KE metrics]

1) $\Rightarrow$ $\exists\, \lambda : \mathbb{C}^* \hookrightarrow \mathrm{Aut}(Y)$

and embedding $M \hookrightarrow \mathbb{CP}^N$ s.t.

$$Y = \lim_{t \to 0} \lambda(t) \cdot M$$

[Luna slice theorem : action of $SL(N+1)$ on Hilbert scheme near Y can be reduced to the action of $\mathrm{Aut}(Y)$ on a "slice" complementary to the orbit of Y



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2) $\Rightarrow$ $DF(M, \lambda) = 0$

If M is K-stable, then we must have $Y \simeq M$

i.e. M admits a KE metric.

What if we only have ω_t for $t<T$, $T<1$?

Strategy of Chen-Donaldson-Sun:

Instead of smooth continuity method, solve

$$Ric(\omega_t) = t\omega_t + (1-t)[D]$$

$D \subset M$ smooth divisor s.t. $[D] = 2\pi c_1(M)$

[could use $\frac{1}{\ell}[D]$ instead of $[D]$, if $[D] \in 2\pi \ell c_1(M)$]

Geometrically this corresponds to

- $Ric(\omega_t) = t\omega_t$ on $M \setminus D$
- ω_t has conical singularities of cone angle $2\pi t$ along D:

$D \times$ distributional Ricci curvature at tip.

Each ω_t is a GH limit of $\omega_{t,\varepsilon}$

$\omega_{t,\varepsilon}$ are smoothings of ω_t, satisfying

$$Ric(\omega_{t,\varepsilon}) > -\omega_{t,\varepsilon}, \quad \text{non-collapsed},$$

$$\operatorname{diam}(\omega_{t,\varepsilon}) < C \quad (\text{if } t > t_0 > 0)$$